
Central Valley Regional Water Quality Control Board

TECHNICAL MEMORANDUM

TO: David Durette, PE
Senior Water Resource Control Engineer

FROM: Melissa Buciak, PG
Engineering Geologist

DATE: 16 September 2025

SUBJECT: REVIEW OF NITRATE AND SETBACK CONDITIONS FOR INDIAN VALLEY COMMUNITY SERVICES DISTRICT, GREENVILLE WASTEWATER PONDS, WATER QUALITY ORDER 2014-0153-DWQ-R5428, GREENVILLE, PLUMAS COUNTY

I have reviewed case file, including the post-Dixie Fire monitoring results and the October 2019 Report of Waste Discharge (ROWD). The case file and ROWD document the wastewater systems.

Indian Valley Community Services District owns and operates the Greenville wastewater ponds (Facility), which is located east of Greenville, in Section 2, Township 26 North, Range 9 East, MDB&M (Assessor's Parcel No. 004-090-032 and 004-080-019). The Facility consists of five stabilization (treatments) ponds, covering approximately 28 acres. Treatment in ponds consists of natural aeration and chemical decomposition. Treated wastewater leaves the ponds through evaporation and infiltration. The ponds are operated in series with influent entering the first pond, then subsequent ponds as needed.

The Facility has had issues with inflow and infiltration (I/I) since the 2021 Dixie Fire. During post-fire cleanup, damage to wastewater infrastructure (risers and cleanouts) occurred. Prior to the Dixie Fire, it was reported in the ROWD that the Facility discharged approximately 192,500 gallons of wastewater per day and maintained 570 service connections. However reported average daily flows in 2024 were approximately 52,900 gallons per day (gpd) but during the dry season (May through September), average daily flows were approximately 13,200 gpd. A recent survey conducted by the Discharger showed there are only 159 active sewers connections in

Greenville. Per the Onsite Wastewater Treatment Systems Policy (OWTS Policy)¹, a “single-family dwelling unit” is expected to generate an average of 250 gpd of wastewater. Therefore, post-Dixie Fire, it is anticipated that approximately 39,750 gpd of wastewater is discharged to the Facility.

Monitoring well logs from the Facility shows that the uppermost (0 to 20 feet) soils consist of clayey silt, silty clay and silty sand. Groundwater depth is shallow and is estimated to be 5 to 20 feet below ground surface. Based on precipitation data from water years 2014 through 2024, the annual precipitation averages 42 inches; during that time the precipitation ranged from a minimum of 21 inches in 2020 to a maximum of 66 inches in 2017.² Mean evaporation at the Facility is estimated to be approximately 55 inches.³ Topography at the Facility slopes gently southeastward, towards Wolf Creek, the nearest surface water body. According to the ROWD, the wastewater ponds are located within Zone X (areas of 0.2 percent annual chance flooding) and surrounded to the west, north, and east by regulatory floodway of Wolf Creek.

Water supply is provided by Indian Valley Community Services District from two wells (No. 1 and No. 2), located approximately 1.5 miles southwest of the Facility.

POTENTIAL THREATS TO WATER QUALITY

Surface water drainage is to Wolf Creek, an indirect tributary of the Sacramento River. Per the Table 3 of the General Order, the setback requirements for Wolf Creek from the wastewater ponds is 150 feet. Wolf Creek is located approximately 150 feet south of Pond 2. Some existing sites may not comply with the setbacks provided in the General Order. Such noncomplying sites may be permitted under the General Order if nuisance conditions do not result from the noncompliance. The wastewater ponds have been in place for over 30 years. Water quality data collected in 2019 is assumed to reflect a steady-state system; percolating effluent does not appear to adversely affect Wolf Creek.

Completion of the Nitrate Checklist in Attachment 1 of Order 2014-0153-DWQ indicates the following flow and rationale:

A1 Exceed 20,000 gpd? Yes.

Average daily flows in 2024 were approximately 52,900 gpd.

¹ 2023, Water Quality Control Policy for Siting, Design, Operation, and Maintenance of Onsite Wastewater Treatment Systems (OWTS Policy), State Water Resources Control Board, April 18.

² [PRISM Group at Oregon State University](https://prism.oregonstate.edu/explorer/) (https://prism.oregonstate.edu/explorer/)

³ [WUCOLS 2000](https://cimis.water.ca.gov/Content/PDF/wucols00.pdf) (https://cimis.water.ca.gov/Content/PDF/wucols00.pdf)

A2 Shallow Groundwater? No.

Shallow groundwater conditions are defined by the OWTS Policy Table 2, *Minimum Depth to Groundwater and Minimum Soil Depth from the Bottom of the Dispersal System*. According to the ROWD, grading analyses were run on soil collected in July 2019 from well borings EFF-1 and MW-002 at approximately bottom of pond depth of 10 feet below top of the levee. Soil in both samples were classified as a silty clay (CL). The estimated permeability value for the silty clay is 10^{-6} centimeter per second. This lithology would likely yield percolation rates greater than 30 minutes per inch (MPI), which corresponds to a minimum groundwater depth of 5 feet. Groundwater measurements collected in August 2019 from the three monitoring wells showed depths ranging from 9 to 18 feet bgs. Therefore the anticipated percolation rate and depth to groundwater exceeds the minimum requirements in Table 2 and the groundwater is not considered shallow.

Conclusion: A nitrogen effluent limit is not required.

MONITORING REQUIREMENTS

To protect water quality, General Order monitoring requirements will be sufficient. In summary, based on Greenville's wastewater treatment system, staff recommend quarterly reporting:

Wastewater Pond Monitoring: Continuous daily flow rate in gpd. Monthly collection of dissolved oxygen and biochemical oxygen demand in mg/L. Monthly notes on freeboard measurements, odors and berm conditions.

Solids Disposal Monitoring: Report handling and disposal of all solids.

Groundwater Monitoring: Monitor and report groundwater quality on a quarterly basis. Analysis of the data and groundwater flow directions shall be performed at least annually. A reduced monitoring and reporting schedule may be requested once adequate data has been collected to characterize the site.

Quarterly monitoring reports shall be submitted by the **first day of the second month after the quarter ends** (e.g. January-March report is due by May 1st). Annual monitoring reports shall be submitted with the subsequent fourth quarter report due by **March 1st following the monitoring year**.